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RESEARCH PAPER

When AI-Based Agents Are Proactive: Implications for Competence and System Satisfaction in Human–AI Collaboration

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Abstract As the capabilities of artificial intelligence (AI) technologies continue to improve, collaboration with AI-based agents enables users to be more efficient and productive. Not only has the quality of AI-based agents' outcomes increased, but they can now help proactively, and even take over entire work tasks. However, users need to be satisfied with the system to remain motivated to collaborate and engage with AI-based agents. Drawing on self-determination theory, a vignette-based online experiment was conducted that revealed that proactive (vs. reactive) help from AI-based agents leads to a higher loss of users' competence-based self-esteem and thus reduces users' system satisfaction. This effect is moderated by the users' knowledge of AI. Higher (vs. lower) levels of AI knowledge cause a greater loss of competence-based self-esteem through proactive (vs. reactive) help. The findings contribute to a better understanding of help from AI-based agents and provide important implications for managers and designers who seek to enhance human–AI collaboration.

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1 Introduction

Enabled by technological advances in artificial intelligence (AI), the capabilities of AI-based agents have evolved significantly and are becoming increasingly relevant for practice (Dwivedi et al. 2023). They have already surpassed humans in terms of accuracy and performance, demonstrate higher competencies in specific tasks, such as image classification (Fuegener et al. 2022), can act autonomously (Bailey et al. 2019; Baird and Maruping 2021), and are able to handle entire work processes (Strich et al. 2021). Furthermore, the increasing capabilities of AI-based agents have led to the emergence of proactive AI-based agents characterized by their ability to initiate interactions without any requests from users (Baird and Maruping 2021). They are capable of offering help proactively by anticipating users' needs (i.e., proactive help), as opposed to reactive AI-based agents, which help as a reaction to user requests (i.e., reactive help) (Morana et al. 2017; Baird and Maruping 2021; Wenninger et al. 2022). For instance, Tabnine, an AI-based agent specializing in programming, can anticipate its users' programming intentions based on their current input in their editor, such as a function title or a comment, and proactively suggest programming code that realizes these intentions (Tabnine 2023). Proactive AI-based agents can potentially increase users' efficiency in human–AI collaboration by approaching and supporting them. Humans often fail to recognize the need for help, even if seeking help can increase their performance (Yzerbyt et al. 1998; Fuegener et al. 2022). Proactive AI-based agents can overcome this obstruction in human–AI

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collaboration by anticipating their human counterparts' needs and potential difficulties for that they offer help without the need for explicit requests (Kraus et al. 2021; Baird and Maruping 2021). In line with this development, AI-based agents are increasingly perceived not only as passive tools but rather as partners capable of meaningful collaboration (Anthony et al. 2023; Chen and Chan 2023; Dennis et al. 2023). However, collaboration (i.e., working together toward a shared goal) usually takes place in a social setting in which the success of interactions often depends on user perceptions, such as satisfaction (Marks et al. 2001; Dennis et al. 2023).

Recognizing the potential competitive advantages of collaborating with AI-based agents that arise from their increasing capabilities, nearly every fifth German organization is already planning to use AI-based agents such as ChatGPT[1] in the future (Bitkom 2023). However, a representative survey from Continental (2023) revealed that more than 50 percent of German citizens are skeptical or dismissive of using systems based on AI (Continental 2023). While collaborating with AI-based agents has the potential to enhance work processes, their potential in organizational contexts is limited by the reluctance of their potential users. Their success depends on users' willingness to work with AI-based agents, which is strongly related to users' satisfaction with these AI-based agents (i.e., users' system satisfaction) (Briggs et al. 2008; Hsiao and Chen 2022). If users are not satisfied with an AI-based agent, they are less likely to engage with it consistently in the future (Bhattacherjee 2001; Ashfaq et al. 2020; Hsiao and Chen 2022). Therefore, maintaining users' system satisfaction is crucial in order to sustain their motivation to collaborate with AI-based agents (Hsiao 2022; Briggs et al. 2008). However, while the increasing capabilities of AI-based agents can promote user productivity, they can also elicit negative reactions from users (Bar-Or and Meyer 2019). For example, receiving help from competent AI-based agents can lead to a decrease in help recipients' self-esteem and negatively affect their perception of their own competence (Lee et al. 2019; Adam et al. 2023a). This is particularly the case when this help is provided in a proactive rather than reactive manner, whereby the recipient may feel that they are being questioned about their competencies (Lee et al. 2019; Harari et al. 2022; Adam et al. 2023a) .

Considering the growing prevalence and capabilities of proactive AI-based agents at work, there is a critical need to gain a deeper understanding of how collaborating with them affects users in human–AI collaboration to utilize their advantages effectively. Receiving proactive rather than reactive help from AI-based agents has the potential to increase users' performance by overcoming their reluctance to seek help by anticipating their needs and offering help effectively when needed (Bar-Or and Meyer 2019; Fuegener et al. 2022). However, previous research on human–AI collaboration has typically not distinguish between different types of initiation when receiving help from AI-based agents (e.g., Fuegener et al. 2021; Gnewuch et al. 2022; Boyacı et al. 2023), although previous studies have demonstrated that receiving proactive rather than reactive help can negatively affect the perceptions of help recipients (Lee et al. 2019; Harari et al. 2022). Furthermore, those studies primarily focused on how collaborating with AI-based agents improves accuracy and performance without delving deeper into users' perceptions in terms of system satisfaction (e.g., Fuegener et al. 2021; Gnewuch et al. 2022; Boyacı et al. 2023), while users' system satisfaction is an important factor in keeping users motivated to interact with AI-based agents (Bhattacherjee 2001; Ashfaq et al. 2020; Hsiao and Chen 2022). Therefore, it is crucial to investigate how and why the type of help initiation—whether proactive or reactive—affects users' system satisfaction while collaborating with AI-based agents. Additionally, previous studies on human–AI collaboration have demonstrated that certain user characteristics are a critical factor when collaborating with AI-based agents (e.g., Meurisch et al. 2020; Pinski et al. 2024). In particular, these studies have highlighted the importance of users' understanding of the capabilities and applications of systems based on AI (i.e., AI knowledge) when receiving help from AI-based agents (Chiu 2021; Pinski et al. 2023a). Understanding the concepts and functionalities of AI can lead to better decision-making (Long and Magerko 2020), affect users' perception of their own competencies (Yang and Aurisicchio 2021), and increase users' performance when collaborating with AI-based agents (Pinski et al. 2023a). However, recent studies have shown that AI knowledge can also hinder human–AI collaboration by reducing users' intention to use AI-based agents (e.g., Tully et al. 2023). Still, despite the growing need to understand how different types of help initiation affect users' system satisfaction and the recognition of the relevance of AI knowledge in human-AI collaboration, we know little about how receiving proactive rather than reactive help from AI-based agents affects users' system satisfaction and what role AI knowledge plays in this process. Therefore, it is important to address these research gaps by examining the following research questions:

[1] ChatGPT is an AI-based agent developed by OpenAI and is designed to interact with the user in a conversational manner. It is capable of answering questions, writing code, and providing suggestions based on the context of the conversation. Moreover, ChatGPT is able to remember what the user said during the conversation and respond to follow-up questions (OpenAI 2022).

1) How does proactive (vs. reactive) help from AI-based agents affect users' system satisfaction?
2) How does users' AI knowledge influence this effect?

To answer these research questions, we drew on self-determination theory and conducted a vignette-based online experiment with 92 participants proficient in programming who experienced an interaction with an AI-based agent. We show that proactive (vs. reactive) help from AI-based agents leads to a higher loss of users' competence-based self-esteem (i.e., a decline in an individual's confidence in their competencies at work (Craig et al. 2019)) and thus reduces users' system satisfaction. We also find that users' AI knowledge moderates this effect, in that users with higher (vs. lower) levels of AI knowledge experience a more significant loss of competence-based self-esteem.

Our study contributes to information systems (IS) research on human–AI collaboration by providing valuable insights in several important ways. We advance previous research, which has mainly focused on how collaborating with AI-based agents improves accuracy and performance without considering different types of help initiation (e.g., Fuegener et al. 2021, 2022; Boyacı et al. 2023), by investigating how users' system satisfaction is affected when receiving help from AI-based agents proactively rather than reactively. We identified that proactive (vs. reactive) help from AI-based agents can lead to lower user system satisfaction. Furthermore, we find that a loss of competence-based self-esteem is a determining mediator for this effect. Building on these findings, we highlight the significance of exploring interventions to reduce the loss of competence-based self-esteem and its adverse reactions caused by receiving help from AI-based agents in human–AI collaboration. Furthermore, as part of our study, we distinguish the degree of users' AI knowledge as an essential characteristic influencing users' system satisfaction when receiving proactive (vs. reactive) help from AI-based agents. Finally, we contribute to self-determination theory by introducing the type of help initiation from AI-based agents as a new important design principle that can affect users' psychological need satisfaction when interacting with technology. Thus, our study uncovers valuable insights for organizations that plan to implement AI-based agents and IS researcher, as it highlights the potential negative effect on users' system satisfaction caused by proactive (vs. reactive) help from AI-based agents in human–AI collaboration.

2 Theoretical Background

2.1 AI-Based Agents, Type of Help, and AI Knowledge

An agent in the context of a human–computer interaction is a kind of software system that has the capability to adapt to dynamic environments and to its users (Schleiffer 2005). In the domains of sales, recommendation, or automation (Li et al. 2015; Benlian et al. 2020; Adam et al. 2023b), agents have taken on a vital role in private households and in industry (Maedche et al. 2019; Mirbabaie et al. 2021; Dwivedi et al. 2023). This development appears to be driven by AI, leading to a shift from simple-reflex agents to learning agents (Russell and Norvig 2016; Stelmaszak et al. 2024). While simple-reflex agents sense inputs and act in a pre-defined manner based on their built-in knowledge, learning agents have an extended backend that enables them to learn from experience and update the knowledge base on which they rely to make decisions. Learning agents often use built-in machine models to learn and update their knowledge. However, they can also use pre-defined rules or formulas to facilitate the process (Kühl et al. 2022). In recent years, most agents have become more autonomous and have shown increasing learning capabilities (Schuetz et al. 2020; Berente et al. 2021), which is why we focus only on agents with the ability to learn and to act autonomously (i.e., AI-based agents). Moreover, recent developments in large language models have further changed the dynamic relationship between users and AI-based agents (Chen and Chan 2023; Dwivedi et al. 2023). AI-based agents that rely on large language models are now able to use natural language indistinguishable from that of a human to assist in creative tasks, write computer code, and be utilized for brainstorming (Chen and Chan 2023; Memmert and Tavanapour 2023; Peng et al. 2023). Overall, these AI-based agents are increasingly perceived not only as passive tools, but rather as partners capable of meaningful collaboration (Anthony et al. 2023; Chen and Chan 2023; Dennis et al. 2023Hao et al. 2024). This development has led to a growing interest in understanding the collaboration between users and AI-based agents and its consequences for user perception (Ariani 2024; Adam et al. 2024a).

One of the key benefits of AI-based agents is their ability to provide help to users based on different types of initiation, including reactive and proactive help. Reactive help from AI-based agents is assistance that is provided by an agent when the user explicitly requests it (Li et al. 2015; Baird and Maruping 2021). Reactive help can appear in the form of answering questions, providing recommendations, or suggesting solutions to problems (Kraus et al. 2021). In contrast, proactive help from AI-based agents is assistance

that is provided by an agent without the user explicitly requesting help. This type of help is based on the ability of AI-based agents to anticipate users' needs or preferences and to provide assistance accordingly (Hukal et al. 2019; Baird and Maruping 2021; Adam et al. 2024b). Proactive help can provide recommendations, remind users of important events or tasks, or notify users of potential problems (Kraus et al. 2021). While the concept of and the differentiation between reactive and proactive help in human–human collaboration is already well established in the management literature (e.g., Spitzmuller and Van Dyne 2013; Lee et al. 2019; Parker et al. 2019; Harari et al. 2022), most of the previous IS literature has made no real distinction between these different types of help initiation (e.g., Fuegener et al. 2021, 2022; Gnewuch et al. 2022), even though reactive and proactive help has differences in terms of design, implementation, and the provided outcomes (Parker et al. 2010). In IS research, reactive help is predominantly welcomed and favorably evaluated by users (e.g., Komiak and Benbasat 2006; Qiu and Benbasat 2009). However, previous research has demonstrated that reactive help can have disadvantages compared to proactive help. For example, Lee et al. (2019) showed that receiving proactive rather than reactive help at work can undermine help recipients' competence and self-esteem. Furthermore, they demonstrated that help recipients showed less gratitude to the helper when they received proactive rather than reactive help (Lee et al. 2019). Additionally, Harari et al. (2022) demonstrated that receiving proactive help was less likely to be accepted than receiving reactive help, as proactive help can challenge the competencies of the help recipient, especially when the help provider has a higher status at work.

Nevertheless, the receipt of proactive rather than reactive help may confer certain advantages. For instance, Maniktala et al. (2023) showed that users who received proactive help from an AI-based agent had significantly higher accuracy in solving problems than users who only had reactive help available to them. Furthermore, Bar-Or and Meyer (2019) demonstrated that users who received proactive rather than reactive help when interacting with technology could increase their task performance, especially when the task was difficult. However, they also showed that users were less interested in receiving proactive help compared to reactive help in the future, although proactive help increased their task performance (Bar-Or and Meyer 2019). Moreover, recent IS research has demonstrated that humans tend not to ask for help from AI-based agents, even if asking for help increases overall task performance (Fuegener et al. 2022). According to Fuegener et al. (2022), a lack of meta-knowledge can explain this phenomenon. Users incorrectly assess their competence compared to the competence of the agent who is providing the help. Consequently, the number of times users request help falls short of the optimal frequency (Yzerbyt et al. 1998; Fuegener et al. 2022). Implementing proactive help can contribute to overcoming this effect by increasing the frequency of users accepting help from AI-based agents (Kraus et al. 2021) and subsequently improving their performance (Fuegener et al. 2021, 2022). However, to take full advantage of the potential performance gains from proactive AI-based agents, it is important to ensure that users are willing to interact with them. Therefore, it is essential that users are satisfied with the AI-based agent because if users are not satisfied with an AI-based agent, they will be less motivated to interact with it (Briggs et al. 2008; Ashfaq et al. 2020; Hsiao and Chen 2022).

Recent literature demonstrates that the effectiveness of receiving help from an AI-based agent can depend on the user's level of AI knowledge (Pinski et al. 2023a). AI knowledge describes the skill level or level of understanding a human has about the capabilities and applications of AI-based systems (Chiu 2021; Pinski et al. 2023a). As AI-based agents have become more integrated into various aspects of daily life and industries, it has become essential for users to have a foundational understanding of the capabilities and applications of AI-based agents. Insufficient knowledge about the capabilities of AI-based agents can lead to user misunderstandings, while higher levels of AI knowledge can help users make better decisions when receiving help from AI-based agents (Long and Magerko 2020). For instance, recent research has shown that a higher level of AI knowledge improves users' ability to make decisions when deciding whether to delegate tasks to AI-based agents or perform them on their own (Pinski et al. 2023a). Despite its potential to increase users' decision-making in human–AI collaboration, knowledge about the capabilities of AI-based agents can also be an obstacle. Previous research has shown that knowledge about the capabilities of a collaborator can negatively affect the perception of one's own competence, especially when the collaborators' capabilities are perceived to be high (e.g., Alicke et al. 1997; Boissicat et al. 2012). As a result, a lower perceived competence can have a negative impact on ones' motivation and well-being (Deci and Ryan 2008). When collaborating with AI-based agents a high level of AI knowledge can hinder the effective utilization of them. In particular, AI knowledge can lead to a lower intention to actually collaborate with AI-based agents (Pinski et al. 2023a; Tully et al. 2023). However, the impact of AI knowledge linked to the different types of help—proactive and reactive—from AI-based agents is still poorly understood.

2.2 Self-Determination Theory in Human–AI Collaboration

While receiving proactive help from AI-based agents can be beneficial in human–AI collaboration (Bar-Or and Meyer 2019; Fuegener et al. 2022), it can also lead to adverse reactions compared to reactive help (Adam et al. 2023a). When receiving help from proactive AI-based agents, help recipients can feel dissatisfied with the system (Wenninger et al. 2022). Individuals can feel challenged in their competencies when receiving proactive rather than reactive help, especially for work-related tasks (Lee et al. 2019; Harari et al. 2022). According to Calvo et al. (2020), self-determination theory can explain the undermining effect on users' competence when collaborating with AI-based agents. Self-determination theory proposes that individuals have three innate psychological needs—autonomy, competence, and relatedness—that affect their well-being and experience (Deci and Ryan 2000). These needs are "essential for ongoing psychological growth, integrity, and well-being" (Deci and Ryan 2000, p. 229). Thereby, autonomy denotes a sense of being in control of one's own behavior (DeCharms 1968; Deci and Ryan 1985). Competence refers to the feeling of being capable and effective in one's engagement with the environment, while relatedness pertains to an individual's sense of being connected to others and having a sense of belonging (White 1959; Deci and Ryan 1985; Ryan and Deci 2000; Deci et al. 2008). When these needs are satisfied, individuals tend to be more engaged and experience a greater sense of well-being (Deci and Ryan 2000; Ryan and Deci 2000). Conversely, when these needs are unmet, individuals may experience disengagement or ill-being (Ryan and Deci 2000). Among other things, external events can contribute to the dissatisfaction of these psychological needs by undermining an individual's sense of autonomy or competence (Deci and Ryan 2012; Ryan and Deci 2000). For instance, when individuals encounter an event that transmits a feeling of being controlled, their need for autonomy can be thwarted. Similarly, events that give individuals feedback can affect their need satisfaction. While positive feedback will increase individuals' perceived competence, negative feedback will transmit the feeling of being incompetent, dissatisfying their psychological need for competence (Deci and Ryan 2012).

In the context of IS, self-determination theory has already been established as a relevant theory for understanding human behavior and perception when using technology (e.g., Karahanna et al. 2018; Lu et al. 2021; Croitor et al. 2022). Croitor et al. (2022) drew on self-determination theory to explain how the perception of input control elements on online platforms can affect user satisfaction. Lu et al. (2021) showed that gamified mechanisms on online forums can motivate users to participate, and Karahanna et al. (2018) demonstrated that the satisfaction of psychological needs can affect users' behavior on social media. Although all three psychological needs can be affected when interacting with AI-based agents (De Vreede et al. 2021; Yang and Aurisicchio 2021), in this study, we focus primarily on the psychological need for competence, as it is particularly relevant in task-oriented contexts (Ryan and Deci 2000; Deci et al. 2008). Furthermore, IS research demonstrates that users' psychological need for competence can be undermined when receiving help from AI-based agents and can be affected by users' AI knowledge (e.g., Calvo et al. 2020; Yang and Aurisicchio 2021). In particular, recent literature shows that collaborating with AI-based agents can affect users' psychological need for competence and thus users' engagement and well-being in human–AI collaboration in various ways (e.g., De Vreede et al. 2021; Yang and Aurisicchio 2021). De Vreede et al. (2021) found that users' psychological need for competence can impact their satisfaction when collaborating with an AI-based agent. Yang and Aurisicchio (2021) demonstrated that users' knowledge of the capabilities of AI-based agents while interacting with them can affect their need for competence, and Nguyen et al. (2022) showed that the satisfaction of users' competence can increase users' system satisfaction when interacting with an AI-based agent. However, while previous research based on self-determination theory has provided valuable insights into how collaboration with AI-based agents can affect individuals, we have little understanding of how and why proactive help (vs. reactive help) from AI-based agents may affect users' feeling of being competent and thus decrease users' system satisfaction in human–AI collaboration.

3 Research Model and Hypothesis Development

Drawing on self-determination theory as our theoretical lens, we derive our research model that sheds light on proactive (vs. reactive) help from AI-based agents and its effect on users' system satisfaction in human–AI collaboration. As depicted in Fig. 1, our research model proposes that proactive (vs. reactive) help from AI-based agents affects users' system satisfaction (H1) and that a loss of competence-based self-esteem mediates this effect (H2). In addition, we theorize that users' AI knowledge influences the indirect effect of proactive (vs. reactive) help from AI-based agents on users' system satisfaction via a loss of competence-based self-esteem (H3).

Prior studies on human–AI collaboration have demonstrated that receiving help from AI-based agents can enhance efficiency and increase users' productivity (e.g.,

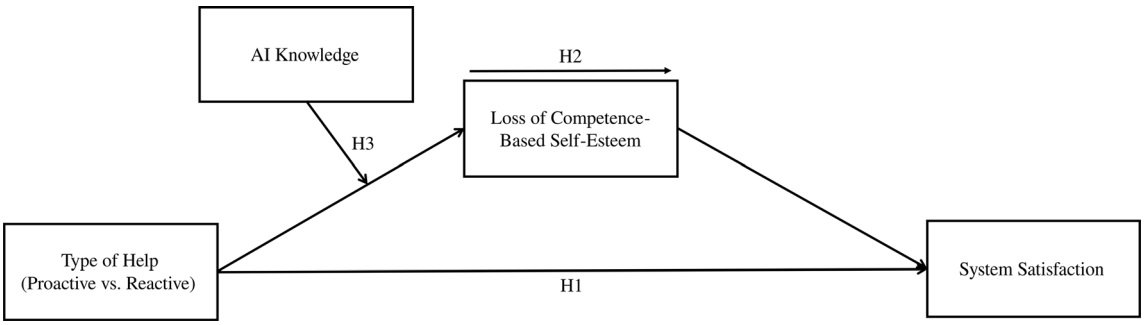


Fig. 1 Research model

Peng et al. 2023). However, previous research has also shown that help from AI-based agents can lead to adverse reactions, especially when help is offered proactively rather than reactively. For instance, Wenninger et al. (2022) demonstrated that receiving help from AI-based agents can reduce user satisfaction, even if the help received was beneficial. Furthermore, Fuegener et al. (2022) found that when users receive proactive help from AI-based agents in human–AI collaboration, some users tend to adjust their behavior to avoid receiving such help. This could result from the fact that users of AI-based agents must be satisfied with them to maintain their motivation to interact with the AI-based agent (Briggs et al. 2008; Ashfaq et al. 2020; Hsiao and Chen 2022). This potential negative effect of receiving help from AI-based agents on user satisfaction is particularly apparent when AI-based agents act autonomously by proactively engaging with their users (Wenninger et al. 2022), as humans tend to prefer non-autonomously-acting AI-based agents (Latikka et al. 2021). In particular, receiving help proactively rather than reactively can negatively affect the perceptions of the help recipients (Lee et al. 2019; Harari et al. 2022; Adam et al. 2023a). All in all, these studies suggest that receiving help proactively rather than reactively may affect users' system satisfaction. Accordingly, we derive our first hypothesis:

H1 Proactive (vs. reactive) help from AI-based agents leads to lower users' system satisfaction.

To further investigate the potential effect of proactive (vs. reactive) help from AI-based agents on users' system satisfaction and to explore its underlying mechanisms, we draw on self-determination theory. Self-determination theory proposes that individuals have a psychological need for competence, referring to the feeling of being capable and effective in their engagement with the environment (Deci and Ryan 2000). When the psychological need for competence is not satisfied, individuals may experience lower self-esteem and satisfaction (Myers and Diener 1995; Reis et al. 2000; De Vreede et al. 2021). Dissatisfaction with the psychological need for competence can arise through interactions with technology (Croitor et al. 2022; Gagné et al. 2022). Self-determination theory states that external events can contribute to the dissatisfaction of the psychological need for competence by conveying a sense of incompetence, undermining individuals' feeling of being capable and effective in their engagement with the environment (Ryan and Deci 2000; Deci and Ryan 2012). In the context of receiving help, especially proactive help can lower the self-esteem and competence of help recipients by transmitting a feeling that their competencies are being questioned, thus decreasing their well-being (Lee et al. 2019; Harari et al. 2022). In particular, receiving help from colleagues at work can negatively affect the competence-based self-esteem of the help recipient (Deelstra et al. 2003).

Therefore, we argue that proactive help from AI-based agents poses a more significant level of dissatisfaction with users' psychological need for competence than reactive help. By acting proactively, AI-based agents transmit the sense that users' competencies are being questioned, leading to a loss of competence-based self-esteem. This unfulfillment of users' psychological need for competence will result in negative feelings, especially lower satisfaction, toward the system that provides help to the user.

H2 Proactive (vs. reactive) help from AI-based agents leads to lower users' system satisfaction through a higher loss of users' competence-based self-esteem.

Individuals' psychological need for competence can be negatively affected by external events, giving them negative feedback relating to their competencies when engaging with their environment (Ryan and Deci 2000; Deci and Ryan 2012). However, the undermining effect of proactive help from AI-based agents on individuals' need for competence depends on how much the external event is perceived as meaningful negative feedback (Deci and Ryan 2012). Thus, users' evaluation of help when seeking or receiving help from another individual is related to their perceived competencies (Bamberger 2009; Harari et al. 2022). Furthermore, Harari et al. (2022) showed that

proactive help challenges individuals' competence, especially when the help provider has a higher status than the recipient (Harari et al. 2022). The capabilities of AI-based agents continue to improve and have become more advanced in various organizational domains (Dwivedi et al. 2023). They have already advanced to the point that they can take over entire work processes, including central tasks employees perform in their professions (Strich et al. 2021). Moreover, AI-based agents surpass the performance of human beings in tasks such as image classification (Fuegener et al. 2022). Users with higher levels of AI knowledge are especially aware of this development and the current capabilities of AI-based agents. While AI knowledge has many advantages in collaboration with AI-based agents, such as improved decision-making, higher AI knowledge levels can reduce the intention to use (Pinski et al. 2023a) and the receptivity of AI-based agents (Tully et al. 2023).

Therefore, we argue that users with greater AI knowledge are more likely to be aware of its technical capabilities and potential to perform better than humans. Consequently, they may perceive the competence-challenging characteristics of proactive (vs. reactive) help from AI-based agents as a more significant impediment to their psychological need for competence. Thus, we hypothesize that the more knowledgeable users are about AI, the more they will perceive proactive help from AI-based agents as meaningful negative feedback on their competencies in engagement with the environment. Therefore, and in line with H1 and H2, we derive our third hypothesis:

H3 The effect of proactive (vs. reactive) help from AI-based agents on users' competence-based self-esteem is moderated by the users' AI knowledge, in that a higher level of AI knowledge leads to a greater loss of users' competence-based self-esteem.

4 Research Methodology

4.1 Experimental Design and Procedure

For our study, we decided to focus on software development as our context, especially programming, for several reasons. Recent technological advances in AI have enabled information systems to support humans in writing computer code more efficiently (Kalliamvakou 2022). Along with this development, AI-based agents are gaining popularity among software developers to support them in writing and improving the quality of their code (Baird and Maruping 2021; Kalliamvakou 2022). For instance, Tabnine and GitHub Copilot[2] can be implemented in the most common programming editors to write code for the user based on prompts, stated code lines, or the project's context (GitHub Copilot 2023; Tabnine 2023). Another example of an AI-based agent that can write code for software developers is ChatGPT, developed by OpenAI. ChatGPT, with over 100 million active users (Similarweb 2023), is designed to interact conversationally and to answer or follow user prompts (OpenAI 2022). AI-based agents, such as ChatGPT and GitHub Copilot, do not just write or recommend code; they can also write code documentation, describe the logic in functions, or find errors in existing code (GitHub Copilot 2023; OpenAI 2022). Those advanced capabilities of AI-based agents specialized in programming have the potential to revolutionize the way software developers work in the future. Therefore, software development is a well-fitting and practically relevant context for exploring users' system satisfaction in human–AI collaboration.

To compare the effects of users' system satisfaction when receiving help from AI-based agents in human–AI collaboration, we conducted a vignette-based online experiment with a 2 × 1 design (type of help: reactive vs. proactive) and between-subject treatments using SoSci Survey.[3] The vignette study methodology is a well-established method in IS research for capturing user perceptions when interacting with agents (e.g., Benlian et al. 2020). While online experiments are associated with high internal validity, they often face difficulties in providing external validity (Aguinis and Bradley 2014). Vignette studies (i.e., an experimental survey method based on a description of a real-world scenario or person in which participants are asked to immerse themselves) address these difficulties by providing the opportunity to simulate a realistic scenario within a controlled online experiment (Atzmüller and Steiner 2010; Eifler and Petzold 2019). The potential high level of realism gives vignette studies higher external validity (i.e., the generalizability of findings to real-world problems) than other experimental methods conducted in a controlled environment while still maintaining high internal validity (Atzmüller and Steiner 2010; Aguinis and Bradley 2014; Eifler and Petzold 2019). However, the design of a vignette study still has the potential to lack external validity, especially compared to non-experimental settings or field experiments, which are typically characterized by their high external but low internal validity (Eifler and Petzold 2019). We chose the vignette study

[2] GitHub Copilot, developed by GitHub Inc., is an AI-based pair programmer based on the OpenAI language model Codex (GitHub Copilot 2023).

[3] SoSci Survey is an advanced tool for creating online questionnaires and experiments (Leiner 2024).

method for our experimental design due to its high internal validity, resulting from a high level of control in the experimental setting while ensuring external validity within its limits (Aguinis and Bradley 2014; Eifler and Petzold 2019). In particular, recent research shows that vignette studies are an effective way to compare the effects of receiving reactive versus proactive help (e.g., Harari et al. 2022). The associated high level of control provides the opportunity to control external factors that may influence the reliability of an experiment that compares the effects of reactive and proactive help, such as the timing when help is asked for or offered. Especially, a reactive help condition requires a high level of control to ensure that the help recipient requests help. In other experimental settings with higher external but lower internal validity, this may not be guaranteed because participants could opt not to ask for help during the experiment. In addition, the survey-based nature of a vignette study is particularly useful for capturing subjective data, such as participants' satisfaction with the system when working with an AI-based agent, whereas other methods in a non-experimental setting are more suitable for measuring objective data, such as performance.

According to established standards for vignette-based online experiments (Atzmüller and Steiner 2010; Aguinis and Bradley 2014), we instructed participants to report their emotions after assuming the role of a fictional person and engaging in a simulated collaboration with an AI-based agent. The experimental procedure of our vignette-based online experiment was as follows. First, we welcomed our participants to the experiment and told them they would be participating in an online survey about technology use at work. We then randomly assigned our participants to one of our two experimental conditions (proactive and reactive help) by means of the SoSci Survey randomization function to ensure randomization with roughly equal distribution across both groups independent from other variables. Next, we asked all participants to step into the shoes of our fictional person, Alex, a software developer at an IT company whose main activity is writing code. In addition, they read that Alex identifies as a code writer. Then, they read that the manager was asking Alex to write code for an important new feature. In the next step, all participants read about the recent introduction of CoCreator™ by the manager. CoCreator™ was presented to the participants as an AI-based agent possessing the same level of coding proficiency as Alex and other human team members so as to avoid unintended effects due to skill differences and to focus the study on the effect of receiving proactive (vs. reactive) help from AI-based agents rather than on performance differentials between help recipients and help providers. After introducing Alex and CoCreator™, the participants experienced one of our two manipulations of *Type of Help* (reactive vs. proactive) according to their assigned condition. For our manipulation, the participants read about and saw a screenshot of the collaboration with CoCreator™. For the sake of realism, we decided to integrate our manipulation into a mock-up of the integrated development environment (IDE) Visual Studio Code. According to a survey by Stack Overflow, Visual Studio Code is the most common IDE among developers (Stack Overflow 2023). See Appendix A (available online via http://link.springer.com) for the exact wording and presentation of the text-based version of our manipulation for both conditions.

To improve the perception of the collaboration, the participants also experienced the manipulation in a 30-s video. In both conditions, Alex begins to type the code for the code-writing task for the new feature and realizes it takes longer than usual to write the code. To avoid any unintended effects of the typed code for the code-writing task, we chose to use a dummy code that appeared realistic but had no specific functionality, as presented in Figure A.1 in the Appendix. Subsequently, CoCreator™ offers help either proactively or reactively. In the reactive help condition, Alex moves the mouse cursor to a button and clicks it to start the collaboration with CoCreator™, in which Alex actively asks for help. In the proactive help condition, CoCreator™ offers help proactively, without any request from Alex. See Figs. 2 and 3 for a detailed presentation of the visualization of our reactive and proactive help manipulation. Both help manipulations began after Alex finished a line of code to reduce the perceptions of disruption. In the last step, all participants answered a post-experimental questionnaire that captured the manipulation checks (i.e., perceived proactivity and reactivity of CoCreator™'s help offer), our constructs, socio-demographics, and several control variables.

4.2 Measured Variables and Manipulation Checks

We measured our dependent variable, *System Satisfaction*, with five items adapted from Brown et al. (2008) (e.g., "I think I could become an enthusiastic user of CoCreator™"). We measured our mediator *Loss of Competence-Based Self-Esteem* with three items adapted from Craig et al. (2019) (e.g., "CoCreator™ makes me feel less confident that I understand things well enough to get work done"). For our moderator variable, *AI Knowledge*, we used four items from Chiu et al. (2021) (e.g., "I know pretty much about AI"). In addition to our constructs for our mediator, moderator, and dependent variables, we measured several control variables to account for potential further explanatory factors in our experiment. In addition to demographics—that is, age and gender (*male, female, and others*)—we measured *Personal Innovativeness* (e.g., "If I

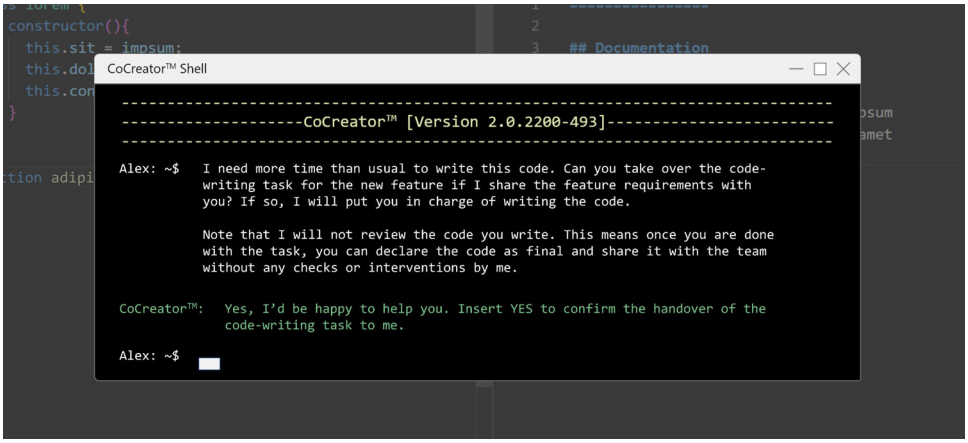


Fig. 2 Screenshot of the reactive help manipulation

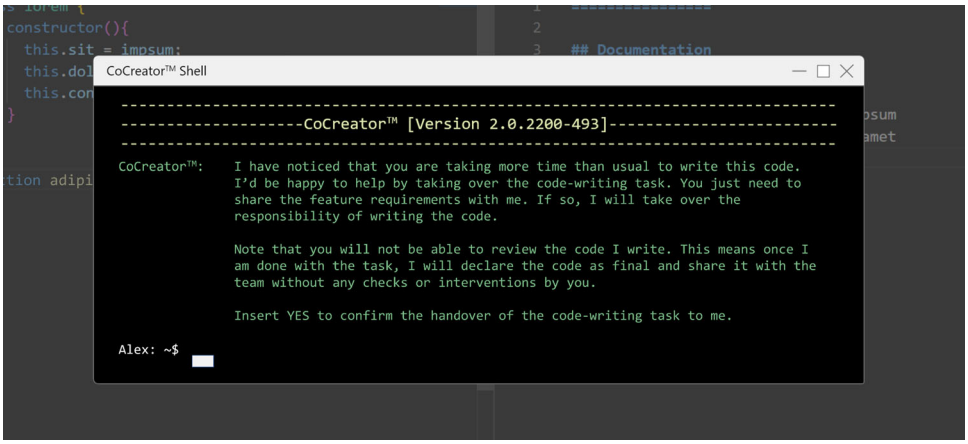


Fig. 3 Screenshot of the proactive help manipulation

heard about a new technology, I would look for ways to experiment with it") (Agarwal and Prasad 1998) to control for users' openness to new technologies, as we assume that users with higher personal innovativeness are less likely to be affected by proactive behavior from AI-based agents, as they are more likely to welcome the initiated interaction. We measured *Product Knowledge* ("Are you familiar with AI pair programmer software, such as GitHub's Copilot?") (Qiu and Benbasat 2010) to control for prior experience in interactions with AI-based agents, as we assume that users who are not familiar with interacting with AI-based agents are more likely to be taken by surprise, affecting their satisfaction when receiving proactive help from an AI-based agent. We measured *Programming Frequencies* ("How often have you written code in the last three months?") (Harari et al. 2022) to address users' experience in programming, as we assume that users' competence in programming could affect users' competence-based self-esteem when receiving help for a programming task from a proactive AI-based agent. Finally, we measured *Negative Affectivity* (e.g., "My feelings are hurt rather easily") (Ayyagari et al. 2011) as it reflects users' "tendency to experience negative emotional states and low self-esteem" (Ayyagari et al. 2011, p. 842), and we assume that users who tend to experience low self-esteem will experience a higher loss of competence-based self-esteem when they receive proactive rather than reactive help from an AI-based agent.

To ensure the reliability and validity of our variables, we measured Cronbach's alpha, composite reliability, and average variance extracted (AVE). For all of our constructs, the values for Cronbach's alpha and composite reliability exceeded the threshold value of 0.70, displaying high internal consistency (Nunnally 1978). In addition, the AVEs were above the threshold of 0.50, indicating good convergent validity, suggesting that our measures are reliable and accurately reflect the underlying construct (Hair et al. 2021). Moreover, the square roots of the AVEs exceeded the inter-construct correlations, providing evidence for the discriminant validity of our constructs (Fornell and Larcker 1981). See Table A.1 and Table A.2 in the Appendix for further information about the measured items.

To check for the effectiveness of our manipulation *Type of Help* (proactive vs. reactive), we measured three items to evaluate the reactive help manipulation (e.g.,

Table 1 Ordinary Least Squares Regression analyses on loss of competence-based self-esteem and system satisfaction

	Loss of competence-based self-esteem		System satisfaction	
	Model (1)	Model (2)	Model (3)	Model (4)
Intercept	2.02	2.20	4.65***	5.22***
	(1.27)	(1.27)	(1.18)	(1.14)
Manipulation				
Type of help	0.86*	0.82*	-1.33***	-1.08**
	(0.42)	(0.41)	(0.38)	(0.38)
Mediation				
Loss of competence-based self-esteem	–	–	–	−0.28**
				(0.10)
Moderation				
AI knowledge	–	−0.23	–	–
		(0.23)		
Type of help x	–	0.70*	–	–
AI knowledge		(0.29)		
Controls				
Age	0.00	−0.01	−0.03	−0.03
	(0.02)	(0.02)	(0.02)	(0.02)
Gender	−0.63	−0.68	0.29	0.11
	(0.44)	(0.43)	(0.40)	(0.39)
Personal innovativeness	−0.10	−0.03	0.27	0.27*(0.13)
	(0.15)	(0.17)	(0.14)	
Product knowledge	0.73	0.77	−0.75	−0.54
	(0.47)	(0.47)	(0.43)	(0.42)
Programming frequencies	0.09	0.07	−0.19	−0.16
	(0.12)	(0.12)	(0.11)	(0.11)
Negative affectivity	0.27*	0.30	0.07	0.15
	(0.12)	(0.12)	(0.11)	(0.11)
R^2	0.14	0.20	0.18	0.26

Table 2 Direct and indirect effects of (moderated) mediation analyses

		AI Knowledge*	**Effect**	**SE**	**CI**	**Significance**
H1: direct	Type of Help—> System Satisfaction	–	−1.08	0.38	[−1.83, −0.33]	Yes
H2: indirect	Type of Help—> LOCBSE—> System Satisfaction	–	−0.24	0.17	[−0.65, −0.01]	Yes
H3: Index of moderated mediation	Type of Help: AI Knowledge—> LOCBSE—> System Satisfaction	–	−0.20	0.12	[−0.47, −0.01]	Yes
H3: Conditional effect	Type of Help—> LOCBSE	3.33	−0.12	0.58	[−1.27, 1.03]	No
		4.67	0.81	0.41	[0.01, 1.62]	Yes
		6.00	1.75	0.55	[0.65, 2.85]	Yes
H3: Conditional indirect effect	Type of Help—> LOCBSE—> System Satisfaction	3.33	0.03	0.17	[−0.34, 0.36]	No
		4.67	−0.23	0.16	[−0.63, −0.00]	Yes
		6.00	−0.49	0.27	[−1.14, −0.07]	Yes

LOCBSE = Loss of competence-based self-esteem; CI = 95% confidence interval; Significance is determined by whether the CI includes zero; *16th, 50th, and 84th percentiles of the distribution, Johnson–Neyman significance region > 4.66

"CoCreator^{TM} agreed to do things for me when I asked") and three items to evaluate the proactive help manipulation (e.g., "CoCreator^{TM} demonstrated initiative in helping me in advance of being asked") adapted from Harari et al. (2022). The results of several ANOVAs indicate that our manipulation achieved its desired effect. The participants in the reactive help condition perceived the help offered by the AI-based agent as more reactive compared to the proactive help condition ($F = 130.10$, $p < 0.001$). Additionally, the participants in the proactive help condition perceived the help offered by the AI-based agent as more proactive compared to the reactive help condition ($F = 49.88$, $p < 0.001$). For more details about the manipulation checks, see Table A.3 in the Appendix.

4.3 Sample Description and Data Collection

We recruited 100 participants with programming experience from the crowdsourcing platform Prolific.co. Prolific.co is an established platform for acquiring subjects for academic studies (Prolific 2023) and is specially designed for scientific purposes (Palan and Schitter 2018). According to multiple studies, Prolific.co respondents' data have demonstrated high reliability and quality (e.g., Peer et al. 2017; Palan and Schitter 2018). To ensure the quality of our data and to capture the population of interest, we restricted access to our study to participants with a minimum age of 18 who were fluent in English and located in the United States. We decided to recruit participants from the United States because participants from the United States tend to provide high data quality on crowdsourcing platforms (Smith et al. 2016). Furthermore, a coherent sample limits potential cultural influences due to different cultural and regulatory contexts. Additionally, we applied filters to include only participants who reported using technology at work at least four times a week and who had experience in computer programming. Finally, we used a filter provided by Prolific.co to obtain participants roughly evenly distributed between the sexes. For our study, we removed eight participants from our dataset due to them failing our attention check (see Table A.1 in the Appendix) (5), for completing the experiment more than twice as fast as the average (2), or for exceeding the maximum allowed completion time by taking eight more hours to finish the survey (1). In total, this led to a final dataset of 92 participants for our statistical analysis. See Table A.4 in the Appendix for details on the descriptive statistics of our final dataset.

To ensure the random assignment of participants to our experimental conditions, we performed several one-way ANOVAs. The results show that there was no statistically significant difference between *AI Knowledge* ($F = 0.92$, $p > 0.05$) and our control variables *Age* ($F = 3.26$, $p > 0.05$), *Gender* ($F = 0.04$, $p > 0.05$), *Product Innovativeness* ($F = 0.22$, $p > 0.05$), *Product Knowledge* ($F = 2.92$, $p > 0.05$), *Programming Frequencies* ($F = 0.36$, $p > 0.05$), and *Negative Affectivity* ($F = 0.03$, $p > 0.05$) between our two experimental groups. Therefore, we can assume that the assignment of participants to our experimental conditions was successfully randomized. In addition, we performed a post hoc power analysis using G* Power 3.1 (Faul et al. 2009) to verify that our final sample size was adequate for our statistical analysis. The power analysis showed that our sample size of $N = 92$ was sufficient, with a power level of 0.81, thus exceeding the recommended threshold of 0.80 (Ellis 2010).

5 Results

To test our hypotheses, we conducted four distinct ordinary least squares (OLS) regression analyses, as depicted in Table 1 (model (1)–(4)). We executed linear regressions on our mediator *Loss of Competence-Based Self-Esteem* and our dependent variable *System Satisfaction*. *Age*, *Gender* (with female coded as one and other as zero), *Negative Affectivity*, *Personal Innovativeness*, *Programming Frequencies*, and *Product Knowledge* (with "Yes" coded as one and "No" as zero) were included as control variables for all regressions. Our independent variable *Type of Help* was binary-coded with proactive help as one and reactive help as zero. Model (1) in Table 1 presents the results of an OLS regression, testing whether *Type of Help* has a significant effect *on Loss of Competence-Based Self-Esteem*. Model (2) displays the results of an additional OLS regression analysis in which we extended the regression model used in model (1) to include *AI Knowledge* as an additional variable to test whether *AI Knowledge* has a moderating effect with *Type of Help* on *Loss of Competence-Based Self-Esteem*. Model (3) presents the results of an OLS regression testing whether *Type of Help* has a direct effect on *System Satisfaction*. For the results in model (4), we extended the regression model used for the results in model (3) to include *Loss of Competence-Based Self-Esteem* as an additional variable, in order to test whether *Loss of Competence-Based Self-Esteem* has a direct effect on *System Satisfaction*. The results of the OLS regression in model (1), (2), and (4) indicate whether *Loss of Competence-Based Self-Esteem* mediates the effect of *Type of Help* on *System Satisfaction* (Shrout and Bolger 2002).

Our results from the third analysis (model (3)) show a significant effect of *Type of Help* (proactive vs. reactive) on *System Satisfaction* ($M = 3.14$ vs. $M = 4.23$, $\beta = -1.33$, $p < 0.001$), indicating support for H1. In addition, our results from the first and fourth analyses show a significant

effect of *Type of Help* on *Loss of Competence-Based Self-Esteem* ($M = 4.21$ vs. $M = 3.45$, $\beta = 0.86$, $p < 0.05$, model (1)) and a significant effect of *Loss of Competence-Based Self-Esteem* on *System Satisfaction* ($\beta = -0.28$, $p < 0.01$, model (4)). These results provide initial support for H2, indicating a significant mediating effect of *Loss of Competence-Based Self-Esteem* on *System Satisfaction* ($\beta = -0.28$, $p < 0.01$). To enhance the robustness of our analysis, we conducted a bootstrap analysis with 5,000 bootstrap samples and 95% confidence intervals (CIs) using PROCESS[4] model 4 (see Figure B.1 in Appendix B) (Hayes 2022). A bootstrap analysis can increase the robustness of mediation analysis with a small- to medium-sized sample by using a resampling technique to determine whether the sampling distribution of the mediated effect differs from zero (Shrout and Bolger 2002). As displayed in Table 2 and in support of H1, the direct effect of *Type of Help* (proactive vs. reactive) on *System Satisfaction* is significant (direct effect = − 1.08, standard error = 0.38, 95% CI = [− 1.83, − 0.33]). Specifically, participants in the proactive help condition were, on average, 1.08 points (on a 7-point Likert scale) less satisfied with the system than participants in the reactive help condition. Thus, proactive (vs. reactive) help from AI-based agents leads to lower user system satisfaction. In support of H2, the results provide evidence that the mediating effect of *Loss of Competence-Based Self-Esteem* on *System Satisfaction* is significant (indirect effect = − 0.24, standard error = 0.17, 95% CI = [− 0.65, − 0.01]). Specifically, participants were, on average, 0.24 points less satisfied with the system as a result of the effect of a loss of competence-based self-esteem when receiving proactive rather than reactive help from AI-based agents. Thus, proactive (vs. reactive) help from AI-based agents leads to lower user system satisfaction via an increased loss of competence-based self-esteem.

Next, we extended our model with *AI Knowledge* as a moderator to test our third hypothesis (H3), suggesting that the effect of proactive (vs. reactive) help from AI-based agents on *Loss of Competence-Based Self-Esteem* and thus on *System Satisfaction* is moderated by *AI Knowledge*. The results of our second analysis (model (2)) indicate a significant interaction between *Type of Help* and *AI Knowledge,* with respect to *Loss of Competence-Based Self-Esteem* ($\beta = 0.70$, $p < 0.05$), providing initial support for H3. Consistent with our prior analysis, we conducted a bootstrap analysis with 5,000 bootstrap samples and 95% CIs using PROCESS model 7 (see Figure B.2 in Appendix B) (Hayes 2022) to provide a more robust moderated mediation analysis. As displayed in Table 2 and in support of H3, the results provide evidence that the moderated mediating effect of *Type of Help* and *AI Knowledge* on *System Satisfaction* via *Loss of Competence-Based Self-Esteem* is significant (index of moderated mediation = − 0.20, standard error = 0.12, 95% CI = [− 0.47, − 0.01]). The negative indirect effect of proactive (vs. reactive) help on *System Satisfaction* via a loss of *Competence-Based Self-Esteem* (H2) increases by 0.20 points (on a 7-point Likert scale) per-point increase in *AI Knowledge*. Thus, the effect of proactive (vs. reactive) help from AI-based agents on users' system satisfaction via a loss of competence-based self-esteem is larger when users have a higher level of AI knowledge. Furthermore, the results in Table 2 show that this effect is only significant when users' AI knowledge is higher than 4.66 on a 7-point Likert scale. Participants with *AI Knowledge* of 4.66 (or 6.00) were, on average, 0.23 (or 0.49) points (on a 7-point Likert scale) less satisfied with the system as a result of the effect of a loss of competence-based self-esteem when receiving proactive rather than reactive help from AI-based agents. In addition, the effect sizes displayed in Table 2 show that *AI Knowledge* had such a large effect on the direct effect of *Type of Help* on *Loss of Competence-Based Self-Esteem* that participants with *AI Knowledge* equal to or greater than 6.00 (on a 7-point Likert scale) showed a direct effect more than twice as large as the average (1.72 compared to 0.82 on a 7-point Likert scale). Therefore, our results show that users' AI knowledge is an important factor that can multiply users' loss of competence-based self-esteem and, thus, system satisfaction when receiving proactive rather than reactive help from AI-based agents.

Figure 4 demonstrates the slopes of the moderation of *AI Knowledge* at different levels (two, one, and zero standard deviations (SDs) below and above the mean) (McCabe et al. 2018), giving an overview of the effects of *Type of Help* on *Loss of Competence-Based Self-Esteem* at different levels of *AI Knowledge*.

To further increase the robustness of our study, we ran several additional analyses as robustness checks (see Appendix B for more information). Specifically, to validate the derivation of our research model, we tested whether our moderator *AI Knowledge* had any additional moderating or direct effects that were not included in our research model. As displayed in Appendix C, we did not find any additional direct or moderating effects of *AI Knowledge* on our mediator, *Loss of Competence-Based Self-Esteem,* or on our dependent variable, *System Satisfaction,* a fact that supports the adequacy of our research model. Furthermore, we conducted an additional mediation analysis grounded in our research model but with *Intention to Use* as an alternative dependent variable, which is highly correlated with *System Satisfaction* (Sun et al. 2014; Hsiao and Chen 2022). We collected *Intention to Use* as an additional

[4] PROCESS is an established tool for OLS regression-based path analysis (Hayes 2023).

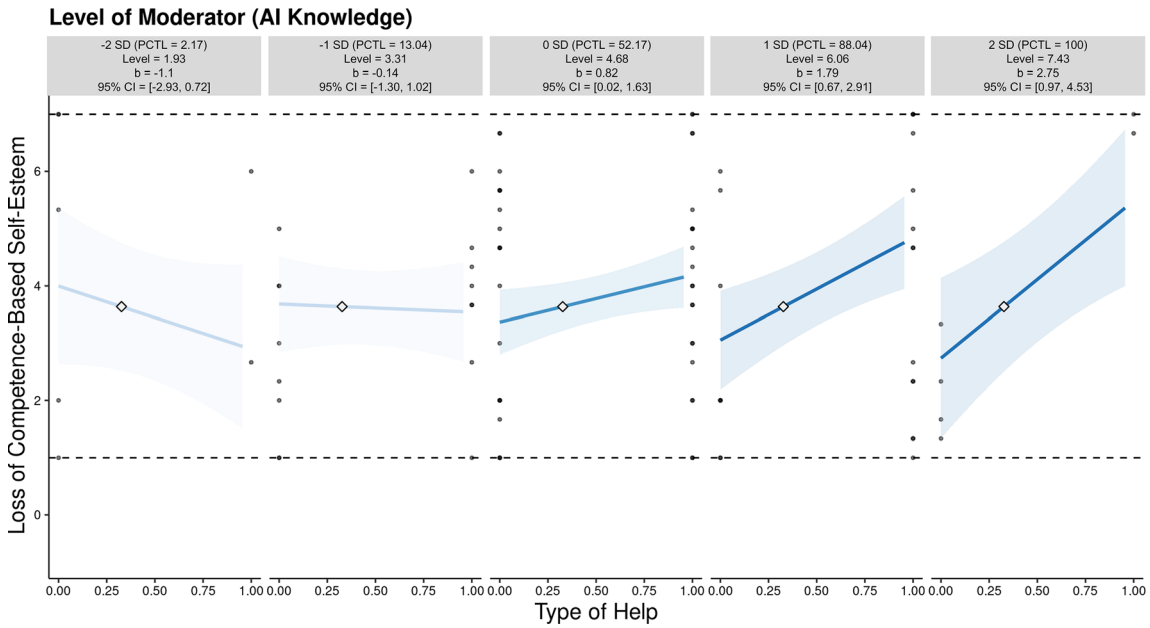


Fig. 4 Illustration of the moderation of AI knowledge on the effect of type of help on loss of competence-based self-esteem; SD = Standard Deviation, PCTL = Percentile, b = Unstandardized Coefficient of the Slope, CI = Confidence Interval

measure during the vignette study. As shown in Appendix B, this analysis yielded similar patterns of results to our main analysis. This consistency across different but related constructs further strengthens the reliability of our findings.

6 Discussion

Collaboration with AI-based agents at work is becoming increasingly relevant for various tasks. This development has evolved due to the increasing capabilities of AI-based agents (Dwivedi et al. 2023). The quality of their outcomes has improved; they can now help proactively, and even take over entire work tasks (Baird and Maruping 2021; Strich et al. 2021). These emerging capabilities of AI-based agents have the ability to increase users' performance. In particular, receiving proactive rather than reactive help from AI-based agents can increase users' performance by anticipating users' needs and offering help without being asked for it (Bar-Or and Meyer 2019; Fuegener et al. 2022). However, recent IS studies show a tendency for users to be deterred when receiving proactive help from AI-based agents (Bar-Or and Meyer 2019; Fuegener et al. 2022). As such, a deeper understanding of how users will experience proactive compared to reactive help from AI-based agents in human–AI collaboration has become more crucial. Therefore, this study aimed to shed light on users' system satisfaction when an AI-based agent offers proactive (vs. reactive) help at work.

Drawing on self-determination theory, we find that users show, on average, about 1.08 points (on a 7-point Likert scale) lower system satisfaction when receiving proactive (vs. reactive) help from AI-based agents for tasks at work in human–AI collaboration (supporting H1). This significant shift in system satisfaction is particularly important because introducing a proactive rather than a reactive AI-based agent to support users at work may be more beneficial in terms of productivity (Bar-Or and Meyer 2019; Fuegener 2022). However, the associated lower likelihood of system satisfaction may result in the benefits of proactive agents not being fully realized, as users must be satisfied with the AI-based agent to be motivated to interact with it (Ashfaq et al. 2020; Hsiao and Chen 2022). Moreover, and in support of H2, we find that this effect is mediated by a higher loss of competence-based self-esteem. In particular, and in support of H3, we find that users with high levels of AI knowledge show, on average, up to more than twice as much (about 0.49 vs. 0.24 points on a 7-point Likert scale) lower system satisfaction through a loss of competence-based self-esteem when receiving help from AI-based agents proactively rather than reactively. However, users with low levels of AI knowledge do not experience a significantly higher loss of competence-based self-esteem and thus do not experience lower system satisfaction due to this effect when receiving proactive rather than reactive help from AI-based agents. These findings have important practical implications as they suggest that designing AI-based agents to be proactive rather than reactive would be less convenient for users to work with,

especially in use cases where users typically have high AI knowledge. In summary, our research findings provide a theoretical foundation on which organizations can build when planning to implement proactive AI-based agents to fully realize the potential of such agents. Therefore, they have significant implications for the nascent field of human–AI collaboration, both in its theoretical and practical aspects, which we will discuss further in the following sections.

6.1 Theoretical Contributions

Our study contributes to IS research on human–AI collaboration in several ways. First, we enhance our understanding of how users' system satisfaction is affected when receiving proactive (vs. reactive) help from AI-based agents. Previous IS research has primarily focused on users' interactions with AI-based agents and how collaboration with them can improve efficiency and performance, ignoring user satisfaction as an important prerequisite in human–AI collaboration (e.g., Fuegener et al. 2021; Boyacı et al. 2023), yet users' system satisfaction is an important factor in keeping users motivated to interact with AI-based agents (Briggs et al. 2008; Ashfaq et al. 2020; Hsiao and Chen 2022). Moreover, recent studies on human–AI collaboration have increasingly employed proactive AI-based agents as the basis of their experimental design (e.g., Bauer et al. 2023; Fuegener et al. 2021; Gnewuch et al. 2022; Schemmer et al. 2023). However, these studies have paid little attention to how the collaboration between human and AI-based agents was initiated, and thus they ignored the effects of different types of initiation when receiving help from AI-based agents. For instance, while Fuegener et al. (2021) demonstrate that receiving help from AI-based agents can increase users' performance, they do not consider that the type of help initiation – in this case, proactive help – could have affected their results. Receiving proactive help from AI-based agents can improve users' performance by anticipating users' needs so that their results could differ in a setting in which help from AI-based agents is provided reactively rather than proactively. Overall, we extend our knowledge of the collaboration between humans and AI by improving our understanding of how users' system satisfaction is affected when they receive help from AI-based agents proactively rather than reactively. Consistent with previous research on human–human collaboration (e.g., Lee et al. 2019; Harari et al. 2022), our findings show that the type of help initiation influences users' perceptions when collaborating with an AI-based agent. We especially found that receiving proactive (vs. reactive) help from AI-based agents leads to lower system satisfaction. This is a significant contribution to the field of IS research, which seeks to differentiate between proactively- and reactively-acting IS (e.g., Morana et al. 2017; Baird and Maruping 2021). Furthermore, we give new insights into users' perceptions of human–AI collaboration and demonstrate that the differentiation between different types of help cannot be ignored. Therefore, we suggest that IS researchers in the field of human–AI collaboration should differentiate between the different types of help from AI-based agents, such as reactive and proactive help, to consider their potential different effects on users' system satisfaction.

Second, and relatedly, we increase our understanding of how collaboration with AI-based agents affects users' system satisfaction by identifying competence-based self-esteem as the driving force for this effect. While some studies have already indicated that receiving help proactively from AI-based agents can lead to adverse reactions from help recipients (e.g., Strich et al. 2021; Fuegener et al. 2021; Wenninger et al. 2022), they do not delve deeper to get to the bottom of these effects. For instance, Wenninger et al (2022) demonstrate that receiving proactive help from autonomously acting AI-based agents can negatively affect users' system satisfaction. However, while they provide some theoretical insights into the possible reasons for this effect, they do not empirically investigate potential underlying psychological mechanisms that might explain why users can react negatively to proactive help from AI-based agents. Moreover, many IS studies continue to prioritize the positive implications, such as increased efficiency, of collaboration with AI-based agents (e.g., Boyacı et al. 2023; Fuegener et al. 2022). However, recent calls within the IS research field (e.g., Adam et al. 2024a; Zhang et al. 2024) have emphasized the necessity for a more comprehensive and nuanced understanding of the potential challenges associated with human–AI collaboration. We extend IS research on human–AI collaboration by taking a closer look at potential negative effects depending on the type of help initiation when collaborating with AI-based agents. Furthermore, we explore the reasoning behind the question of why users experience a loss of system satisfaction when receiving help from AI-based agents proactively rather than reactively by falling back on self-determination theory, which posits that feelings of competence are crucial for engagement and well-being (Ryan and Deci 2000). Our findings show that a loss of competence-based self-esteem is a determining factor for lower system satisfaction caused by proactive (vs. reactive) help from AI-based agents. Our reasoning is that receiving proactive help from AI-based agents transmits the feeling that users' competencies are being questioned. This finding is in line with management literature suggesting that receiving proactive rather than reactive help at work can decrease the self-esteem of help recipients and negatively affect their perception of their competence in human–

human collaboration (Lee et al. 2019; Harari et al. 2022). Thus, our study provides valuable insights by demonstrating that future research should consider competence-based self-esteem as an underlying mechanism when evaluating users' reactions to different types of help from AI-based agents in human–AI collaboration.

Third, we identified the degree of users' AI knowledge as an essential characteristic that affects users' system satisfaction by amplifying the loss of competence-based self-esteem caused by proactive (vs. reactive) help from AI-based agents. While most previous research has primarily seen high levels of AI knowledge as a positive characteristic in human–AI collaboration, improving the experience and acceptance of AI-based agents (e.g., Schoeffer et al. 2022; Pinski and Benlian 2024), recent studies have demonstrated that AI knowledge can also lead to adverse effects toward the utilization of AI-based agents by reducing the intention to use them (e.g., Pinski et al. 2023a; Tully et al. 2023). Our study contributes to this recently obtained insight by revealing that high levels of AI knowledge can negatively influence users' perceptions during human–AI collaboration, depending on the type of help initiation when collaborating with AI-based agents. We especially found that users have an even higher loss of competence-based self-esteem and, thus, lower system satisfaction while collaborating with AI-based agents when receiving help proactively rather than reactively. Therefore, we demonstrate that it is crucial also to highlight the negative aspects of increased AI knowledge. Moreover, we suggest that IS researchers should consider users' AI knowledge when evaluating users' perceptions in human–AI collaboration, especially when help is provided via different types of initiation. Thus, this finding is an important and counterintuitive insight, revealing a situation in which a higher level of knowledge can have a negative impact on users' perceptions.

Finally, we contribute to self-determination theory by introducing the type of help initiation from AI-based agents as a new important design principle that can affect users' psychological need satisfaction when interacting with technology such as AI-based agents in human–AI collaboration. While self-determination theory has already been established as an important theory in IS research to derive design principles and to understand users' perception in human–AI collaboration (e.g., De Vreede et al. 2021; Yang and Aurisicchio 2021), previous studies, as far as we know, have not introduced the type of help initiation as an important design principle, affecting users' psychological need satisfaction when interacting with technology (e.g., Pinski et al. 2023b; Yang and Aurisicchio 2021). Therefore, we extend our understanding of what specific design principles can affect psychological needs demonstrating that the type of help initiation from AI-based agents, especially, receiving proactive rather than reactive help from AI-based agents, can affect users' competence-based self-esteem leading to lower system satisfaction. This is an important insight researchers should consider when deriving IS artifacts based on self-determination theory.

6.2 Implications for Practice

Our research findings have practical implications for organizations that use or plan to use AI-based agents to support their employees in their work activities. Our study suggests that receiving help from proactive rather than reactive AI-based agents at work can challenge employees' competencies, leading to lower employee system satisfaction. However, collaborating with proactive AI-based agents can be more efficient than collaborating with reactive AI-based agents, increasing employees' productivity (Bar-Or and Meyer 2019; Fuegener et al. 2022). To effectively handle this trade-off between employees' system satisfaction and productivity when implementing AI-based agents at work, we recommend several actions for designers of AI-based agents and managers.

We recommend that managers should deploy reactive rather than proactive AI-based agents for tasks that are strongly related to the employees' competencies or that do not benefit as much from collaboration with AI-based agents, such as low-time-consuming tasks. This ensures a higher level of system satisfaction for employees when collaborating with AI-based agents, while the unrealized increase in productivity remains within limits. For other tasks, we recommend implementing proactive AI-based agents and introducing measures to counteract the adverse effects of proactive AI-based agents on employees' system satisfaction. As one of these measures, we suggest that managers should gather regular feedback after employees start working with proactive AI-based agents to get information on how satisfied they are when receiving help from those agents. This would give managers the opportunity to counteract potential dissatisfaction quickly. Organizations need to ensure that employees feel valued and respected in their roles and provide opportunities for employees to develop their skills and competencies without receiving help from AI-based agents. Competence-promoting measures can be introduced to satisfy employees' need for competence. Regular positive feedback, competence-enhancing workshops, and trainings in the area of work in which AI-based agents are introduced can help to increase employees' satisfaction of their psychological need for competence (Deci and Ryan 2012). Furthermore, while receiving proactive rather than reactive help from AI-based agents in human –AI collaboration can negatively impact employees' system satisfaction, a deep sense of collaboration could help to mitigate this effect by satisfying their

psychological need for relatedness (Ryan and Deci 2000; Deci et al. 2008). Managers should therefore emphasize the collaborative characteristics of proactive AI-based agents when introducing them. Finally, designers should implement the option to (de-)activate the proactivity of AI-based agents, thus allowing users to have the option to decide by themselves if they want to receive proactive help.

In addition, our results suggest that employees' AI knowledge further amplifies the negative effect on employees' system satisfaction when receiving proactive rather than reactive help from AI-based agents. When employees have a predominantly low level of AI knowledge, managers can expect a lower loss in system satisfaction from their employees when introducing proactive rather than reactive AI-based agents. In contrast, when employees show predominantly high levels of AI knowledge, managers must expect a higher loss in system satisfaction. Therefore, employees' AI knowledge should be considered as an important basis for deciding about implementing proactive AI-based agents or introducing measures to counteract the adverse effects of proactive AI-based agents on employees' system satisfaction. In industries or activities where employees are likely to have higher AI knowledge, such as in software development, managers should rely less on proactive AI-based agents or at least provide a high level of measures, such as competence-enhancing workshops, to maintain employee system satisfaction when implementing proactive rather than reactive AI-based agents. Finally, training and educating employees about AI to help them understand its capabilities and limitations can be beneficial and is recommended in organizational contexts (Pinski and Benlian 2024). However, we suggest that this training and education should also include information about human capabilities and advantages compared to AI to enhance employees' competence-based self-esteem, especially when the implementation of proactive AI-based agents is conceivable in the future.

6.3 Limitations and Directions for Future Research

Despite our theoretical and practical contributions, our study has several limitations that may offer interesting directions for future research. First, our experiment is based on a specific form of assistance from and area of application of AI-based agents—taking over a task at work in the area of software development. In addition to that specific context, AI-based agents are also used in many other contexts, such as medical healthcare and financial services (e.g., Jussupow et al. 2021; Strich et al. 2021). We believe that the context we have selected is relevant and timely, so that our results provide important insights. However, future research could extend our research and strengthen the robustness of our findings by investigating how proactive (vs. reactive) help from AI-based agents can affect users' system satisfaction in other application areas of AI-based agents in human–AI collaboration at work.

In addition, we drew on self-determination theory, particularly on the psychological need for competence, to explain how proactive (vs. reactive) help from AI-based agents affects users' system satisfaction. However, competence is only one of three psychological needs that may affect users when interacting with AI-based agents (De Vreede et al. 2021; Nguyen et al. 2022). In particular, the psychological need for autonomy, along with the psychological need for competence, is particularly important when receiving help from AI-based agents (Calvo et al. 2020; Nguyen et al. 2022). To extend our study and further investigate the reasons for receiving proactive rather than reactive help from AI-based agents, we recommend that future research should extend our study by considering users' perceptions of autonomy and relatedness as an influential variable when receiving proactive (vs. reactive) help from AI-based agents.

Next, our study took place in an experimental setting by conducting a vignette-based online experiment that was limited to a one-time interaction. The methodology of a vignette-based online experiment has high internal validity and is already an established method in the field of IS research for capturing user perceptions when interacting with agents (Benlian et al. 2020). We believe this methodology is an appropriate way to begin exploring the nuanced effects of receiving proactive rather than reactive help from AI-based agents. However, a key limitation of the vignette methodology is the lack of real-world interaction, which may limit external validity. Therefore, our study may not fully capture the complexity of human–AI collaboration in a real-world work environment, where users could behave differently when receiving help from AI-based agents, which would affect the generalizability of our findings. Additional studies could address this limitation and verify the results of our study by building on a method with higher external validity that is grounded on a real work environment instead of a simulation-based vignette study. Furthermore, it is conceivable that as users collaborate with AI-based agents over time, they become accustomed to receiving help from AI-based agents, thereby changing the effects of receiving help from AI-based agents on users' system satisfaction. Therefore, and to deepen our understanding of how users' system satisfaction is affected when receiving help from AI-based agents in human–AI collaboration, we recommend that future research should test our hypothesis in an actual work environment over a more extended period with multiple interactions. In doing so, future research could also account for additional potential moderators, such as the impact of the actual quality of the provided help, users' self-esteem,

and users' competence in solving the related task without receiving help from AI-based agents.

Finally, for our online experiment, we recruited participants with programming experience located in the United States on the crowdsourcing platform Prolific.co. While Prolific.co is a well-established platform for recruiting participants in the field of IS research, it is possible that the participants recruited through crowdsourcing platforms, such as Prolific.co, may not fully represent the general population of users who regularly interact with AI-based agents in work settings. Furthermore, users with programming experience and regular interactions with technology may behave differently when receiving help from AI-based agents than other users. Further research could enhance the generalizability and robustness of our findings by replicating our study with diverse samples and cultural contexts beyond the United States and the context of software development. In particular, a cross-cultural study could offer valuable insights into how cultural differences might influence users' system satisfaction when receiving proactive rather than reactive help from AI-based agents.

7 Conclusion

In conclusion, our study explores the evolving landscape of AI-based agents and their impact on users' system satisfaction in human–AI collaboration. Enabled by technological advances in AI, AI-based agents can increasingly act autonomously, are increasingly able to handle entire work processes, and already outperform humans in specific tasks. This development has led to the emergence of proactive AI-based agents, which are reshaping the way in which humans interact with AI-based agents. Our findings provide essential insights into how help from AI-based agents, offered proactively rather than reactively, affects users' system satisfaction and how users' AI knowledge moderates this effect. However, despite their increasing relevance in practice and research, we believe there is more to be understood about proactive AI-based agents. We have just begun to uncover the vast array of chances and risks that arise when humans collaborate with proactive AI-based agents. More research is necessary to fully comprehend and harness their potential in human–AI collaboration. To this end, our paper provides essential insights that can be built upon in practice and in future research.

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